

CHAPTER 3

Software Requirement Analysis

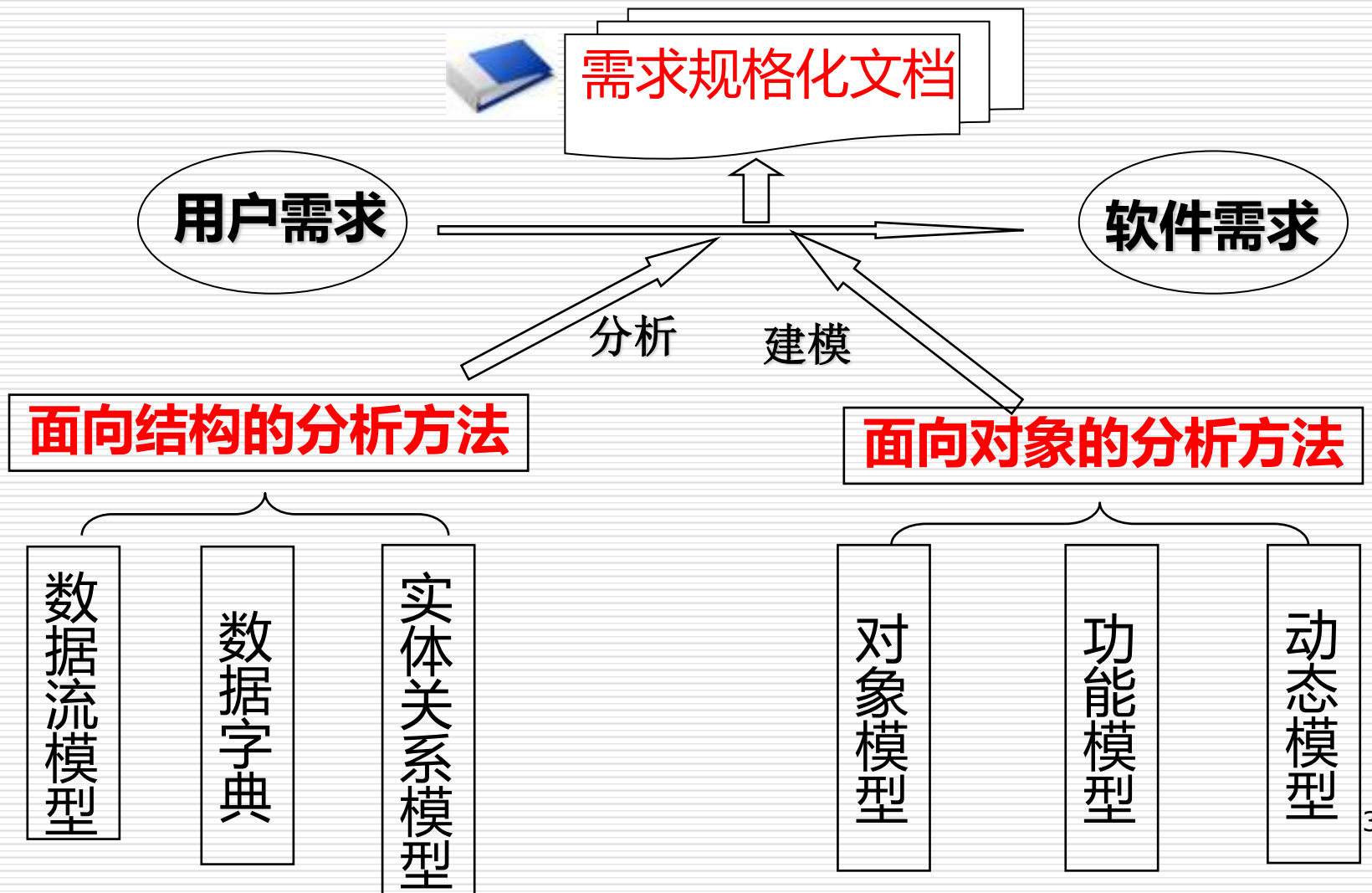
Requirements Engineering

Part 1- Analysis



Part 2- Specification

需求分析的内容



Requirements Specification 目的

- To provide a **representation** of the software for the customer's review and approval
- Developed as a **joint effort** between the developer and the customer
- **Serve as** basis for review for both customer and developer
- **Direct** software design and development
- **Culmination** of requirements analysis

Quality of Requirements Specification

(1) unambiguous

(2) complete

(3) verifiable

(4) consistent

(5) modifiable

(6) traceable

方法

□可混合使用的模型

- ✓ DFD, ERD, DD,...

- ✓ Use case, Class Diagram, Sequence Diagram,.....

- ✓ 其它模型

□使用但不限于面向结构，面向对象方法

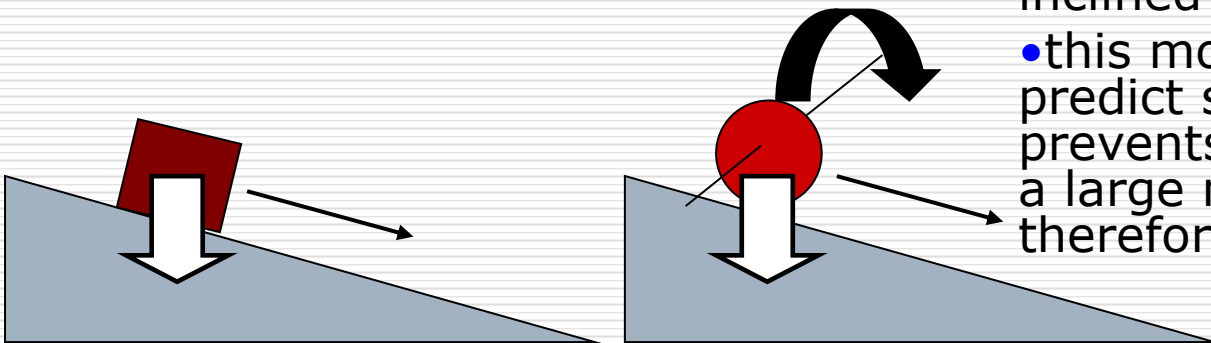
Use other models

□ mathematical models

- many mathematical models are used in engineering or physics
- usually very abstract
- for example: a person who is the passenger in a lift may be abstracted to just their mass when analysing the load in the lift for lift acceleration, braking, motor power, building structural strength

- rolling inertia or friction may be ignored when modelling objects on a slopes as blocks sliding on inclined planes.

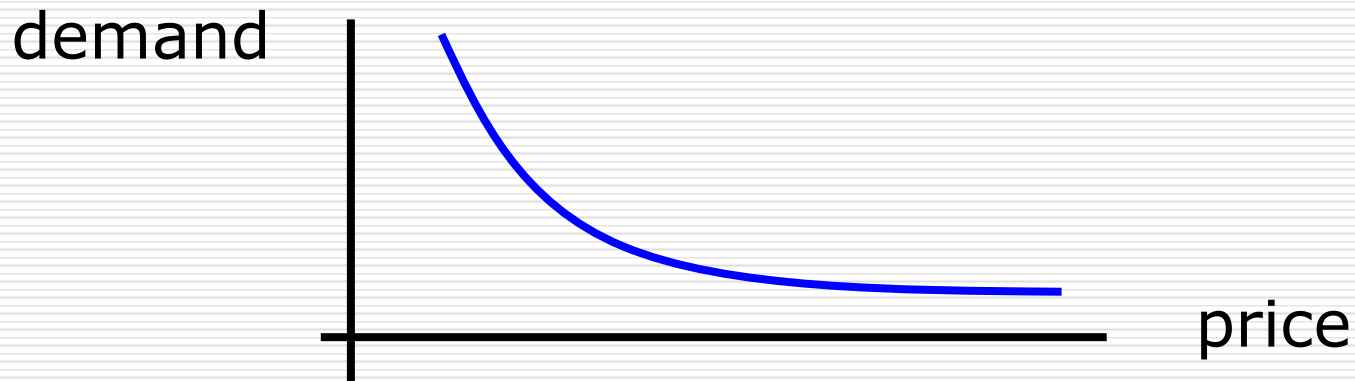
- this model will be inadequate to predict speed and time if friction prevents slipping and the object has a large moment of inertia and therefore will roll more slowly



Use other models

☐ **economics models**

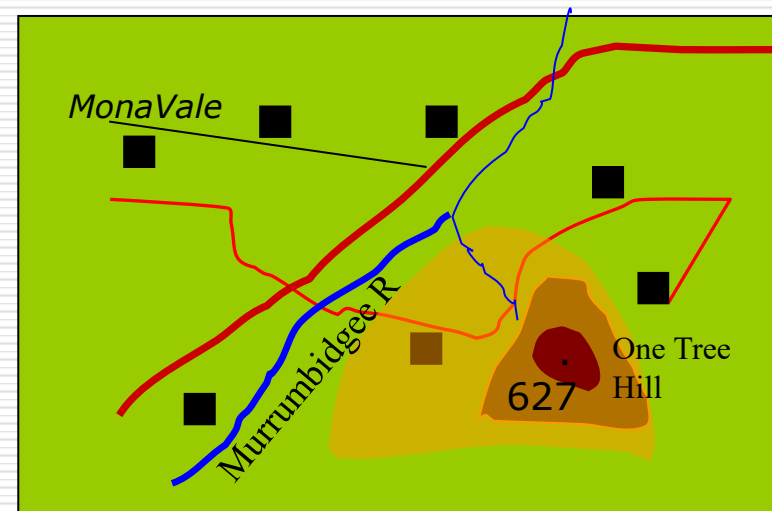
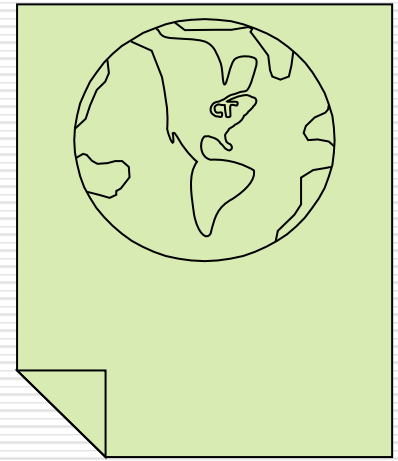
- ☐ price versus demand curve
- ☐ many assumptions to simplify the market – time, season, fashion, whether the product is an unsubstitutatable necessity or a choice



Use other models

□ geographical models

- a map is a model that represents some abstracted aspect of the real world in a diagram: borders of land and sea, political borders, routes of roads (and their intended traffic carrying capacity), heights and vegetation, geology, temperatures



Use other models

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Warnier Diagram

IPO Diagram

Specification 书写模板

1. Introduction

- 1.1 Purpose of the system
- 1.2 Scope of the system
- 1.3 Objectives and success criteria of the project
- 1.4 Definitions, acronyms, and abbreviations
- 1.5 References
- 1.6 Overview

2. Current system

3. Proposed system

3.1 Overview

3.2 Functional requirements

3.3 Nonfunctional requirements

3.3.1 Usability

3.3.2 Reliability

3.3.3 Performance

3.3.4 Supportability

3.3.5 Implementation

3.3.6 Interface

3.3.7 Packaging

3.3.8 Legal

3.4 System models

3.4.1 Scenarios

3.4.2 Use case model

3.4.3 Object model

3.4.4 Dynamic model

3.4.5 User interface—navigational paths and screen mock-ups

4. Glossary

中文需求规格说明书—书写模板

(4) An example project

When to stop requirement analysis

- ✓ requirement specification document finished
- ✓ audit, check
- ✓ change ratio of requirement $<$ a pre-defined value

Course Project

2025-11-06

利用 **Rational Rose** 工具, 完成需求描述的UML分析建模

(1) 共享单车手机APP的需求描述

学号 $\leq$ 2351285

(2) 手机私家车拼车软件系统的需求描述

2351288 $\leq$ 学号 $\leq$ 2352188

(3) 手机公园导游软件系统的需要描述

2352201 $\leq$ 学号 $\leq$ 2353344

(4) 城市公交车无人驾驶系统的需求描述

2353366 $\leq$ 学号 $\leq$ 2353781

(5) 智能衣服电子系统的需求描述

2353798 $\leq$ 学号 $<$ 2399999